

CST8506
ADVANCED
MACHINE LEARNING

Week 1
Introduction

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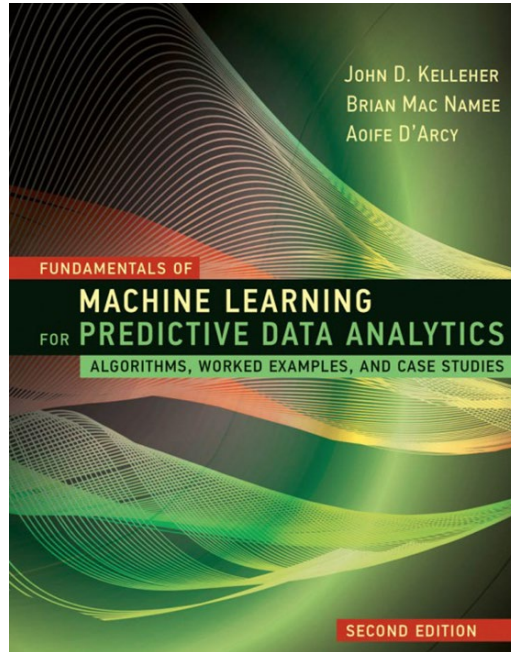
Course Topics

- Preprocessing
- Classification by NN, SVM, Naïve Bayes
- More clustering algorithms
- Class-imbalance problem
- Time Series
- More fusion techniques
- Visualization tools



Required Textbook (Same book as level 1)

Kelleher, J. D., D'Arcy, A., & Namee, M. B. (2020). *Fundamentals of machine learning for Predictive Data Analytics: Algorithms, worked examples, and case studies, Second Edition*. Cambridge, MA: The MIT Press. ISBN: 978-0-262-04469-1



Work

- Lecture: 2 hours
- Hybrid: 1 hour
- Weekly Labs – 2 hours
- Assignments
- Final Project (Mandatory)
- Midterm
- Final exam



Grades

- Final Exam (25%)
- Midterm (15%)
- Assignments (20% = 2 x 10%)
- Labs (20%, 5 labs, equal worth)
- Mandatory Final Project (with Presentation AND report) (20%)

Final project – group work (group of 3 members)



Programming

- Python, RapidMiner
- Visualizations – Python libraries, Power BI, Tableau etc.



- Brightspace – Course Information

